Low Level Design (LLD).

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What is Low Level Design?
Low level design focuses on designing the internal structure of an application by identifying:

1. Classes & objects
2. Their relationships
3. Data flows
4. How DSA solutions fit into this structure.

Difference between DSA & LLD:

⇒ DSA (Data structure & Algorithm):
solves isolated problems like :-
1. shortest path
2. sorting
3. searching (Binary search)
4. Heap, graphs etc.

⇒ Low level design (LLD):

Decide :-
→ What object exist in the system
→ How they interact
→ where Algorithm are applied inside the object structure.

Case study - Building "QuickRide" (Uber like App).

⇒ DSA first Approach :-

steps: ① Represent city as graph. (nodes = intersections, edge = roads).

⇒ use dijkarsta for shortest route
⇒ use min heap for nearest driver matching.

Problems?

⇒ No proper class design (User, Rider, Driver, Payement etc)

⇒ No security Handling (Mobile number masking).

⇒ No Integeration with payement / notijications.

⇒ no scalabul-ity consideration.
(Millions of view on platform / Applications Handling)

Now - LLD Approach:

Step 1:) Identify Entities:-
① User ② Rider ② Driver ③ Location
④ Notijication Service
⑤ Payement service etc.

Step 2:) Define Relationships:-

① User Interact with Rider
② Rider connect to Location
③ Payement Gateway handles Payements
④ Notification service sends updates.

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step3:) Consider Non functional Requirement:-
* scalability
* Data security
* Maintainability.

Step 4:) Apply DSA inside this structure:

↳ shortest path finder
↳ Driver matching heap

Core Principles of LLD:

① **Scalability:** ① should handle millions of users
② Easy to extend features.

② **Maintainability:** ① Easy debugging
② New features shouldn't break old ones.

③ **Reusability:** ① should be loosely coupled modules
eg:- ② generic Notification service. reusable across all Apps.

LLD VS HLD

LLD	HLD
classes, object, relationship, Design pattern, code structure, UML diagram	Tech stacks database choice Load Balancer cost Optimization Autoscaling (AWS/GCP) cloud Deployement (CI/CD).